Unit 10, Lesson 5: Interpreting Graphs of Functions

Benchmark

MA.8.F.1.3 Analyze a real-world written description or graphical representation of a functional relationship between two quantities and identify where the function is increasing, decreasing, or constant.

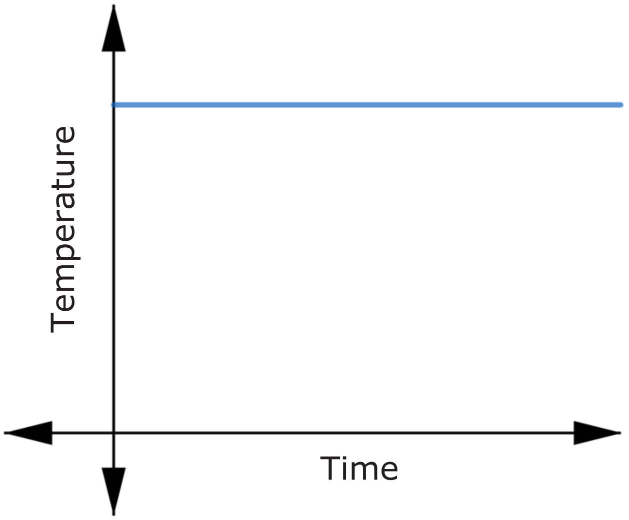
Learning Targets

- ☐ I can identify where a function is increasing, decreasing, or constant from its graph.
- ☐ I can write a description of a graphical representation of a function in context.

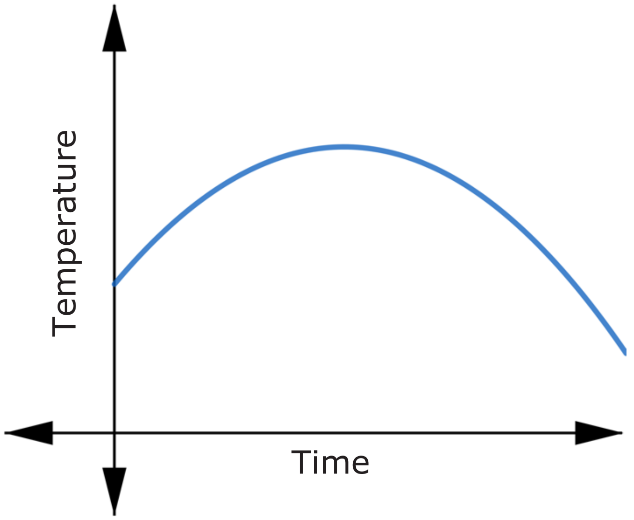
10.5.1 Warm-Up: Which One Doesn't Belong?

Four graphs show the relationship between temperature and time.

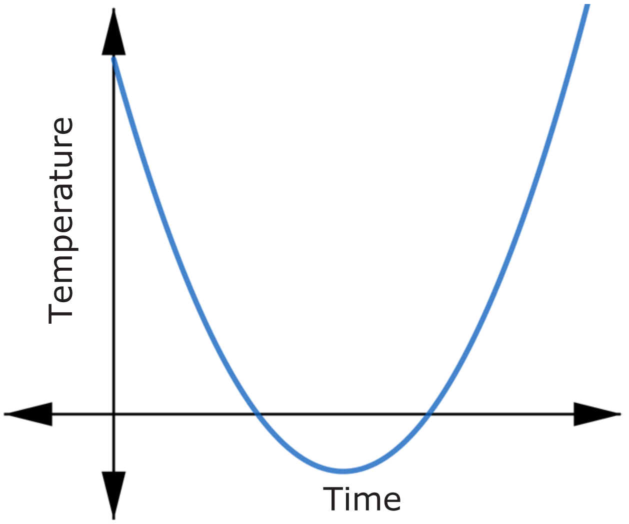
A



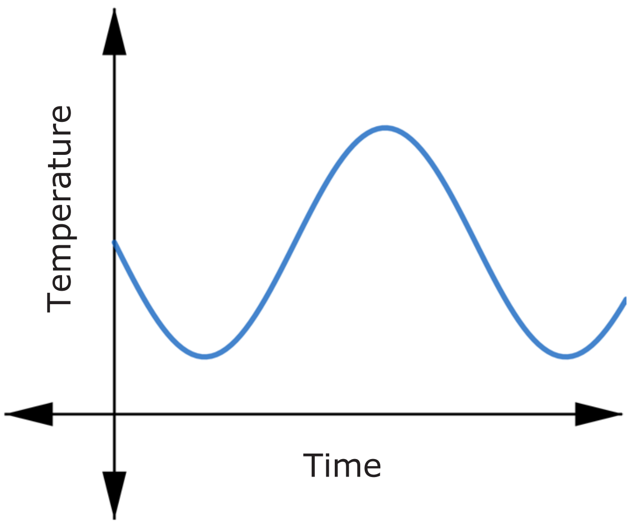
B



C



D

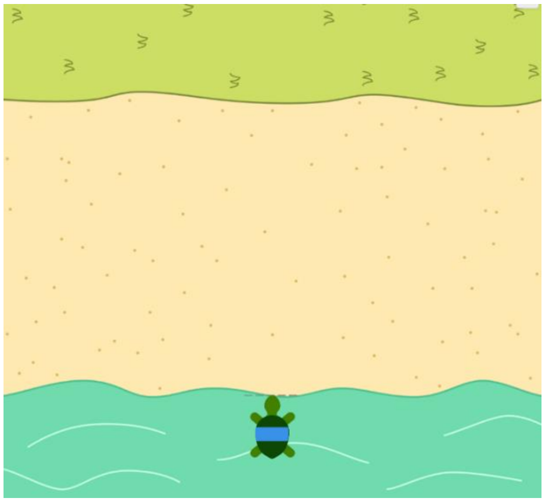


1. Which one doesn't belong?

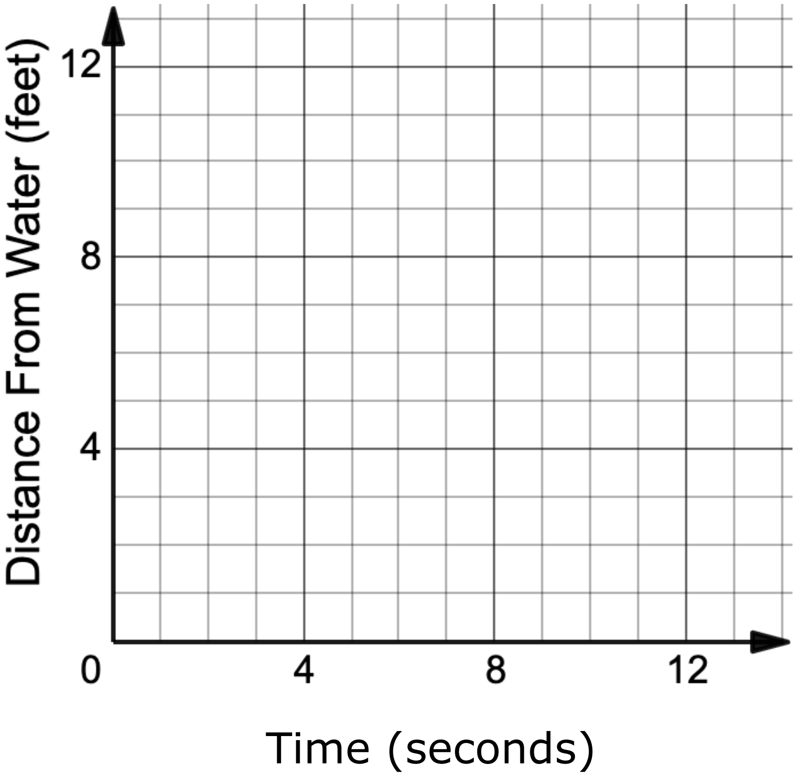
2. Explain your choice.

10.5.2 Exploration: Distance Versus Time Graphs

1. A turtle is traveling across the sand.



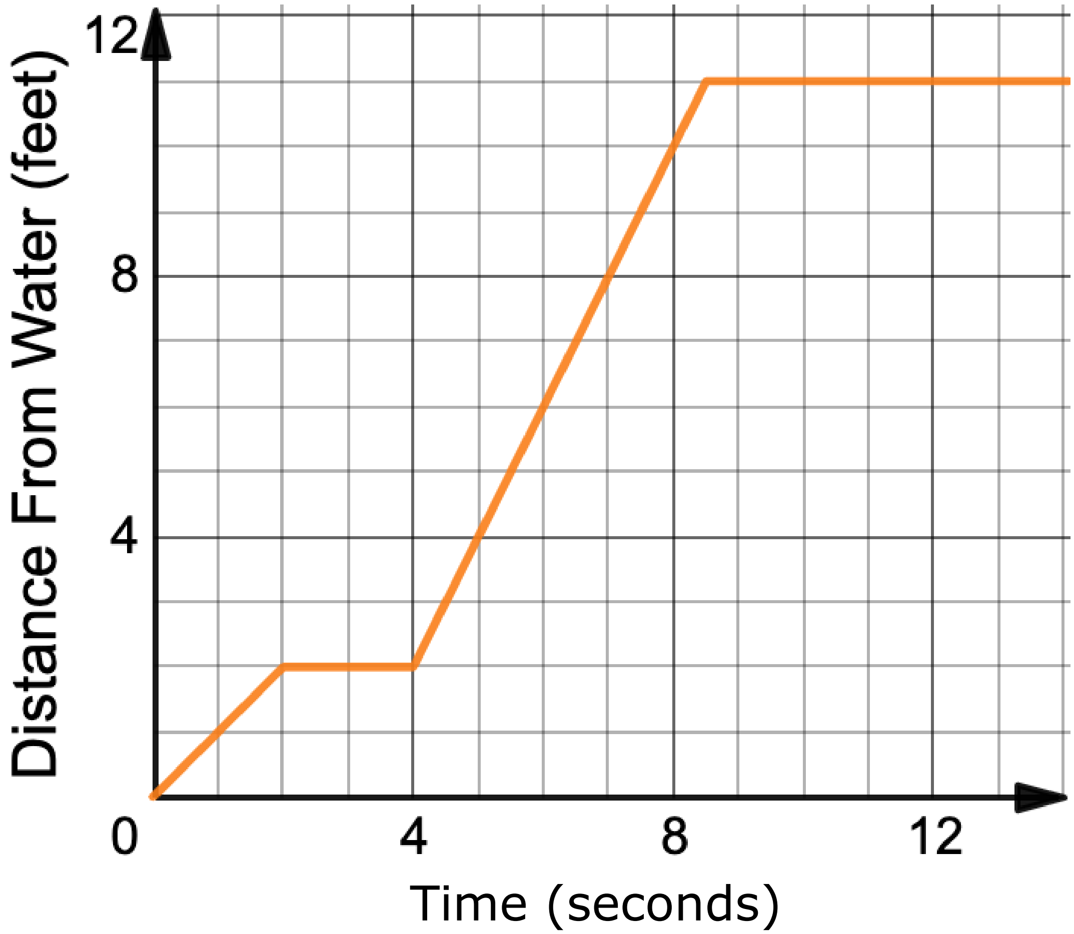
a. Draw a graph using line segments to represent a possible path the turtle could take.



b. Write a brief story describing the turtle's journey based on your graph.

c. Share your graph and story with your partner. Discuss any differences in your graphs or interpretations. Make revisions to your story or graph, if necessary.

2. Luca drew the graph shown to represent a new turtle's journey across the sand.



a. On your own, label the graph as described below.

- Label any parts of the graph where Luca's turtle is walking away from the water with the letter a.
- Label any parts of the graph where Luca's turtle is standing still with the letter b.
- Label any parts of the graph where Luca's turtle is walking toward the water with a letter c.

b. Compare your labels with your partner and resolve any differences, if necessary.

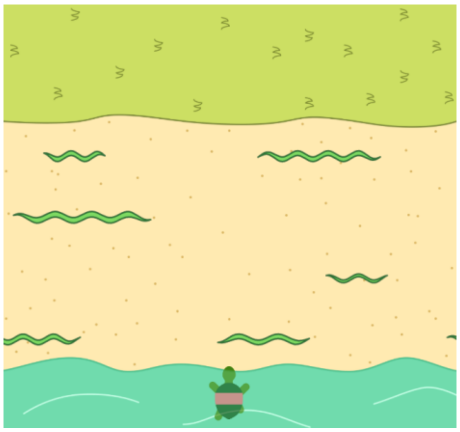
3. Work with your partner to answer the following using Luca's graph.

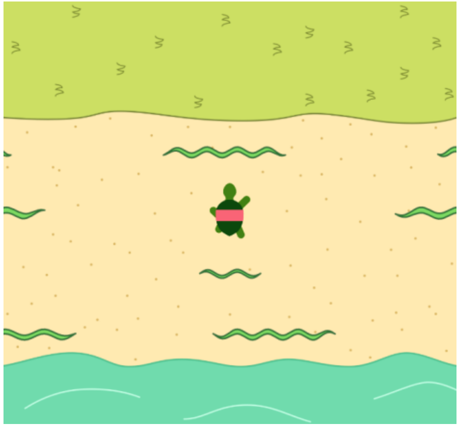
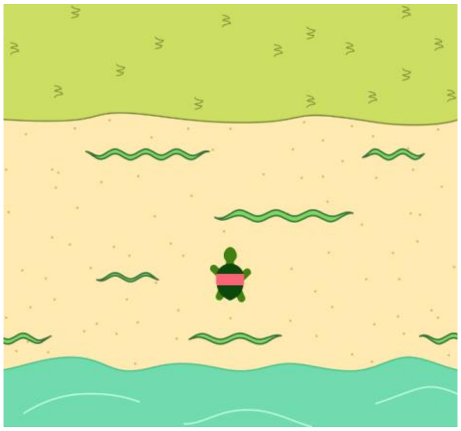
   a. What is the turtle's distance from the water at 8 seconds?

   b. When is the turtle's distance from the water 4 feet?

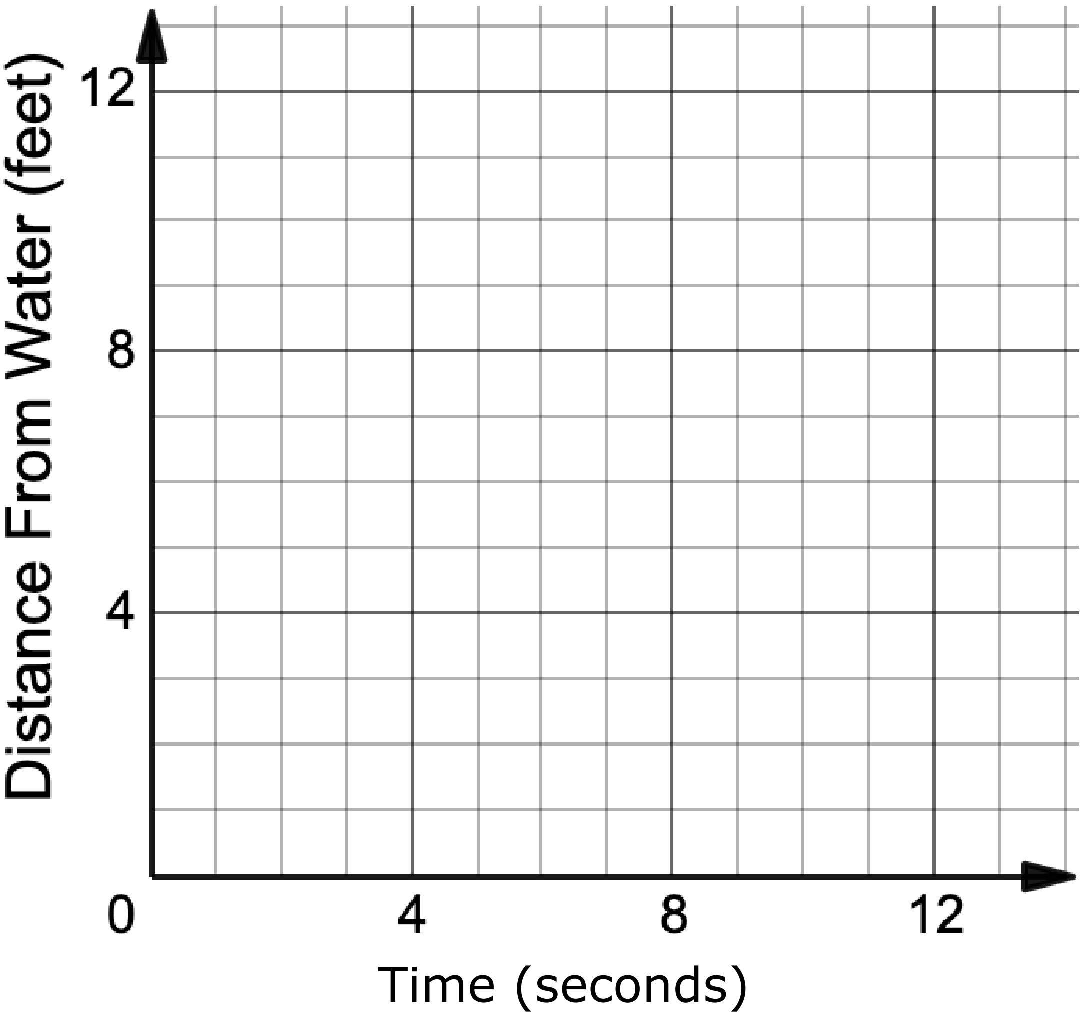
   c. Based on the graph, the turtle is walking between 0 and 2 seconds, and between 4 and 8.5 seconds. Determine whether the turtle is traveling faster between 0 and 2 seconds, or between 4 and 8.5 seconds.

4. Four screenshots and brief descriptions from an animation are shown. In the animation, a turtle is being careful to avoid crawling snakes as it walks across the sand.

Screenshot	**Description**
	The turtle begins in the water.

	Avoiding the crawling snakes, the turtle walks forward at a constant rate until the point shown, where he stops for a few seconds while a snake passes by.
	Then, the turtle backs up to avoid another snake. He waits a few seconds there.
	The turtle then hurries across the rest of the sand to safety.

a. On your own, draw a graph of the turtle’s distance from the water over time based on the description in the table.



b. Compare your graph with your partner. Discuss your graphs and make any changes, if necessary.

10.5.3 Bring It Together: Distance Versus Time Graphs

In graphs comparing time as the independent variable and distance as the dependent variable, the slope of the line represents ________________.

Speed vs. Velocity

Speed is the rate at which an object is moving in units of distance per unit of time.
Velocity is the rate *and* direction of an object's movement.

The slope of a line gives both the rate of change and the direction of the change, positive or negative, so it can be used to determine both the speed and the velocity of an object.

1. Complete each statement.

   a. On a distance-time graph, a horizontal line

      means
      - ○ the object is not moving.
      - ○ the object is moving at a constant speed.

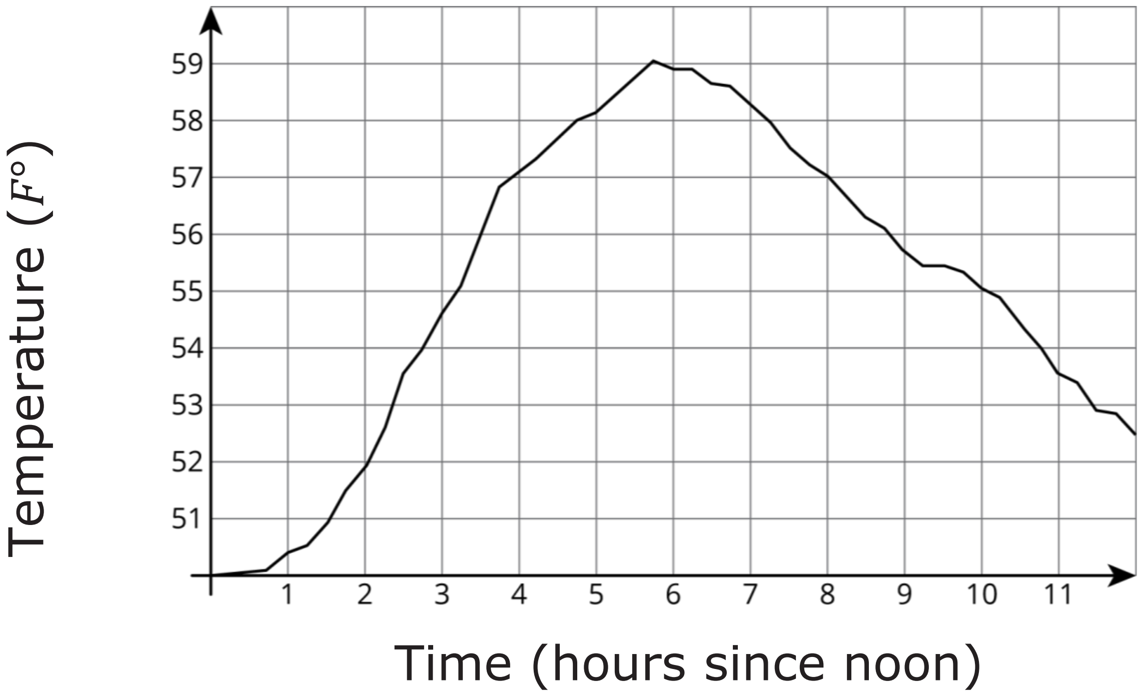
   b. On a distance-time graph, a diagonal line

      means
      - ○ the object is either speeding up or slowing down.
      - ○ the object is traveling at a constant speed.

2. How can the speed of the object be determined for a line segment on a distance-time graph?

10.5.4 Guided Practice: Interpreting Graphs of Functions

1. The graph shows the temperature between noon and midnight one December day in Freeport, Florida.



a. Complete the statement.

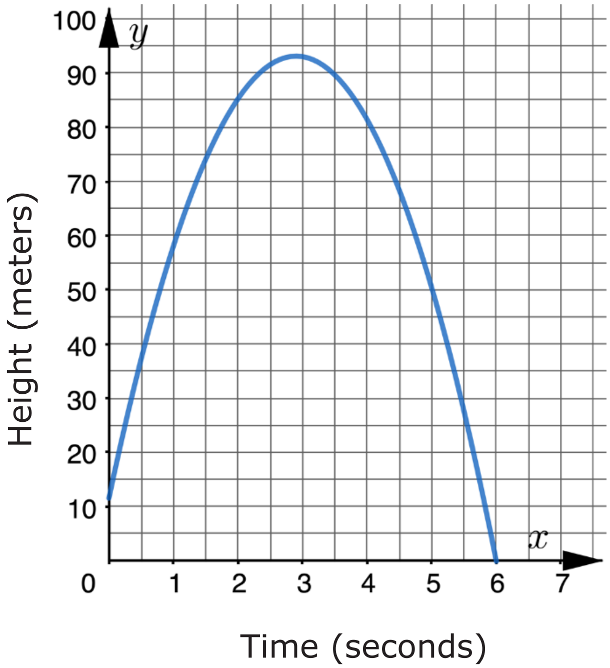
In this relationship, ○ time ○ temperature is a function of ○ time. ○ temperature.

b. Was it warmer at $3:00$ p.m. or $9:00$ p.m.?

c. Approximate the time the temperature was at its highest.

d. Complete the statement.
When the input for the function is 8, the output is ______. This means that at ___$:00$ P.M., the temperature was _____°F in Freeport, FL.

2. The graph represents a bottle rocket that is shot upward from an outdoor stage and then falls to the ground. The independent variable is time, in seconds, and the dependent variable is the object's height above the ground, in meters.



a. Complete the description of the story told by the graph.

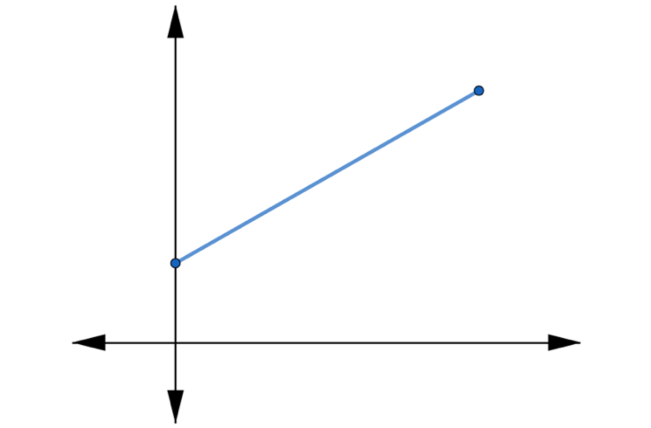
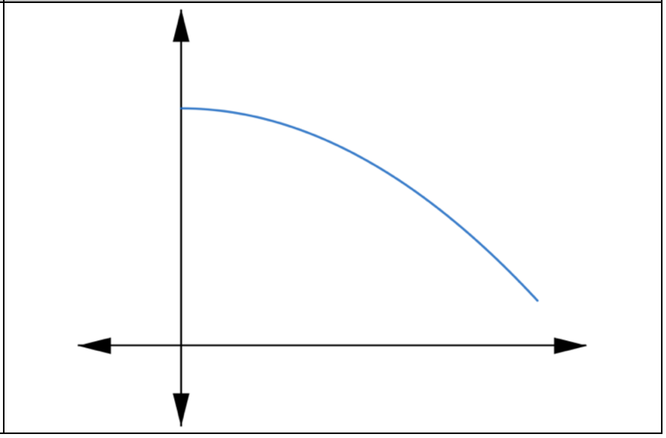
The bottle rocket is launched from an outdoor stage that is ______ meters above the ground. It continues flying upward for about ______ seconds until it reaches approximately ______ meters above the ground. The rocket then begins to fall, landing on the ground after about ______ seconds.

b. Complete the statement and the inequality.
The domain of the function that represents the rocket's path is between ____ and ____ seconds.

$\boxed{} \leq x \leq \boxed{}$

10.5.5 Wrap-Up: Ticket Out the Door

1. Two graphs are shown.
2.

Graph A	Graph B
	

a. Complete the table by writing the letter of the graph that matches each description. Then, name possible independent and dependent variables that could be used to label the axes.

Description	Graph	Independent Variable	Dependent Variable
A plant grows the same amount every week.			
The day started very warm but then it got colder.			

b. Complete the statement.

Graph
- ○ A
- ○ B

shows the function
- ○ decreasing
- ○ increasing

over the interval shown to model how the
- ○ number of weeks
- ○ height of the plant

changes over

the
- ○ number of weeks.
- ○ height of the plant.